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Probing the microscopic flexibility of DNA from melting temperatures

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The microscopic flexibility of DNA is a key ingredient for understanding its interaction with proteins and drugs but is still poorly understood and technically challenging to measure. Several experimental methods probe very long DNA samples, but these miss local flexibility details. Others mechanically disturb or modify short molecules and therefore do not obtain flexibility properties of unperturbed and pristine DNA. Here, we show that it is possible to extract very detailed flexibility information about unmodified DNA from melting temperatures with statistical physics models. We were able to retrieve, from published melting temperatures, several established flexibility properties such as the presence of highly flexible TATA regions of genomic DNA and support recent findings that DNA is very flexible at short length scales. New information about the nanoscale Na^+ concentration dependence of DNA flexibility was determined and we show the key role of ApT and TpA steps when it comes to ion-dependent flexibility and melting temperatures.

DNA has some peculiar and poorly understood flexibility properties. For example, early studies revealed a curious rotation of the purine and pyrimidine rings during stretching[1], which was possibly one of the first signs of its helicoidal structure that would be discovered two years later[2]. Six decades later, many questions remain open, even basic ones such as that of the order of magnitude of the DNA stretching modulus. Experiments that pull long segments of DNA (10^2–10^5 base pairs (bp); for a review, see ref. 3) report a DNA stretching modulus of the order of 1,000 pN. In contrast, for very short length scales (30 bp or less), two completely unrelated experimental techniques, atomic force microscopy[4,5] and the small-angle X-ray scattering interference approach[6], were used to measure these flexibilities and it was found that DNA has a stretching modulus closer to 100 pN. There is however little knowledge on the smallest length scale possible, that is, for nearest-neighbour base pairs. The largest variations in flexibilities are usually found in pyrimidine–purine steps (TpA, that is, a T · A base pair followed by an A · T base pair, and CpG) and purine–pyrimidine steps (ApT and GpC) (as shown by atomic force microscopy[7] and protein-bound DNA (ref. 8)).

Current flexibility measuring techniques always disturb the DNA molecule somehow or measure its conformation while bound to proteins[8] or with nanoparticles attached[4]. In other words, at present there are no flexibility measurements that can be used to probe the DNA in solution without mechanically perturbing it or by attaching something to it. This raises a question: are there other kinds of measurement from which we could obtain further information about the microscopic flexibility of DNA? One possibility is to look at melting temperature experiments. The thermodynamic properties of DNA are largely dependent on flexibility, as molecular vibrations provide ways of storing thermal energy. The prospect of obtaining flexibility properties from melting temperatures is appealing as this is a relatively standard and straightforward experimental technique for which there is already a body of published data available. Furthermore, one would be able to obtain information about the flexibility of DNA without mechanically disturbing it and under solvent conditions that may not be easily accessible to other flexibility measuring techniques.

However, to obtain any flexibility information from DNA melting temperatures we need theoretical models that link both flexibility and thermodynamics properties in a realistic way. At present, the most widely used approach for modelling the flexibility is the use of the worm-like chain (WLC) model, which has been routinely applied to model DNA flexibility since 1972 (refs 9, 10). For long scales, this has been quite successful in reproducing persistence lengths. However, the WLC approach is generally applicable only to polymers that are much longer than their persistence lengths. Therefore, it is not surprising that it has been challenging to apply this model to short-scale measurement[4,5,11]. Crucially, the WLC model is not well suited to answer our question because we cannot associate it with melting temperatures. Statistical physics models such as the popular Poland–Scheraga model provide no simple way either to retrieve flexibility parameters[12]. Within the simplest thermodynamic descriptions of DNA, this leaves us with the Peyrard–Bishop model[13], which does include elastic constants in its basic formulation. Furthermore, the Peyrard–Bishop model separates the contribution of the base-pair hydrogen bond from the nearest-neighbour stacking interaction, that is, it has the two main ingredients of the DNA molecular interactions, which can be retrieved separately. We used this model successfully for the calculation of melting temperatures[14,15]. However, the Peyrard–Bishop models have not been used in the context of DNA flexibility and therefore it is not clear whether their elastic constants are realistically related to DNA flexibility.

Here, we investigate how the elastic constants of the Peyrard–Bishop model relate to DNA flexibility parameters and what new information about DNA flexibility we can obtain in this way. Our main approach to this problem is to calculate melting temperatures of a large set of short sequences for which melting temperatures already exist. We then optimize our model parameters to bring the calculated and experimental melting temperatures into agreement. So far, this apparently straightforward procedure has remained a computationally unfeasible task. Despite the apparent simplicity

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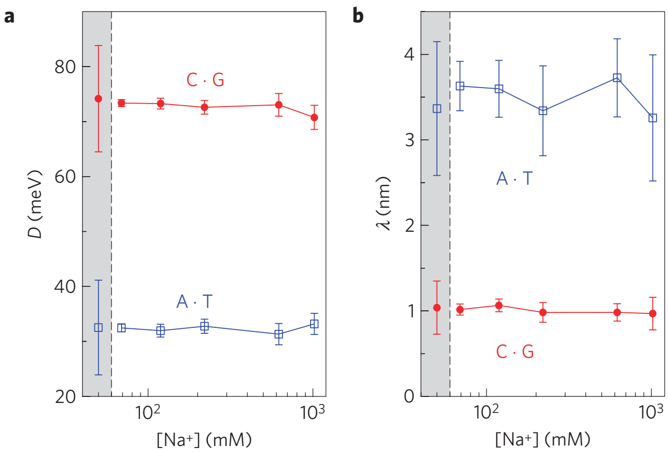

Figure 1 | Optimized Morse-potential parameters. a,**b**, Potential depth D (**a**) and width λ (**b**) as a function of salt concentration. The grey rectangle shows the parameters obtained for the first step of the minimization procedure where we impose a logarithmic salt-concentration dependence given by Supplementary Equation (S3), which are used as starting points for the second step where we minimize for each salt concentration independently. The error bars were calculated using the bootstrap method over 500 modified data sets (see Supplementary Methods).

of the Peyrard–Bishop model, using it for a complete melting temperature calculation is numerically very intensive[16]. To obtain a better theoretical fit for melting temperatures, we need to vary about 14 parameters and calculate melting temperatures for hundreds of sequences. Fortunately, it is now possible to speed up the numerical calculations by several orders of magnitude by using the concept of thermal equivalence that we proposed recently for short DNA sequences[14,15]. This method now places us in a position to carry out the numerical tasks of using the Peyrard–Bishop model to process melting temperature data and work out more realistic parameters, including those related to DNA flexibility. Here, we used published melting temperature data from ref. 17 and we were able to obtain the detailed behaviour of stacking parameters under varying salt concentration. These stacking parameters are associated with the elastic properties of double-stranded DNA. Our results show for instance that the softest sequences correspond to TATA sequences in agreement with this well-known property of DNA flexibility[18–21]. In addition, we obtain the Na^+-dependent flexibility at the base-pair level and show that ApT and TpA are pivotal to understanding DNA flexibility.

The starting point of the Peyrard and Bishop[13] models is a set of two energy terms describing the stacking interaction w and the base-pair interaction V. For instance, for a base pair A·T followed by another C·T, that is, a dimer step ApC, we write the configurational part of the model Hamiltonian as

$$U(\mathrm{ApC}) = w(\mathrm{ApC}) + V(\mathrm{A\cdot T}) \qquad (1)$$

where the dimer step is written as ApC (from the 5′ end of the DNA molecule to the 3′ end)[13,16]. In its simplest form, the Peyrard–Bishop[13] model uses a Morse potential to describe the base-pair interaction, that is, the hydrogen bonds, for example, for an A·T base pair

$$V(\mathrm{A\cdot T}) = D_{\mathrm{A\cdot T}}\left(e^{-y_{\mathrm{A\cdot T}}/\lambda_{\mathrm{A\cdot T}}} - 1\right)^2$$

where y is the relative displacement of the base pair from its equilibrium position, D is the depth of the Morse potential and λ controls the width of the potential. The stacking interaction is modelled as a harmonic potential; for a base pair A·T followed by C·T, this is written as

$$w(\mathrm{ApC}) = \frac{k_{\mathrm{ApC}}}{2}\left(y_{\mathrm{A\cdot T}}^2 - 2y_{\mathrm{A\cdot T}}y_{\mathrm{C\cdot T}}\cos\theta + y_{\mathrm{C\cdot T}}^2\right)$$

where k is an elastic force constant and θ is a twist angle between neighbouring base pairs for which we take a fixed value $\theta = 0.01$ (ref. 15).

In essence, this model separates the nearest-neighbour interactions (stacking interaction w) from pure base-pair interactions V. Therefore, the base-pair parameters in the Morse potential are not context dependent, which means that $V(\mathrm{A\cdot T}) = V(\mathrm{T\cdot A})$ and $V(\mathrm{C\cdot G}) = V(\mathrm{G\cdot C})$, reducing the required number of parameters. Still, for Watson–Crick base pairs, we need ten elastic force constants k and two pairs of Morse potential parameters D and λ, in a total of fourteen parameters.

To calculate melting temperatures, one would typically use the Hamiltonian in a classical partition function to obtain the average displacement of the base pairs over a broad range of temperatures. Close to the melting temperature, this average displacement increases sharply as in a first-order transition[13,16]. Calculating melting temperatures in this way is numerically intensive, requiring up to one hour on a standard 2 GHz PC processor (see also processing times reported in ref. 16). It is not possible, for this type of calculation, to attempt linear regression to extract the unknown parameters as one would do, for example, for Gibbs free energies[22,23]. One way to reverse engineer adequate Hamiltonian parameters from existing experimental temperatures would be to let these parameters vary freely over a certain range until the calculated temperatures are close enough to the experimental data. However, varying a set of 14 parameters for hundreds of sequences is not computationally feasible with usual melting-temperature calculations.

Recently, we developed a new approach to deal with the question of thermal equivalence of DNA (refs 14, 15). In essence, we showed that the thermal properties of a given DNA sequence can be represented by a single parameter without the need for the time-consuming complete temperature calculation[16]. This melting index was shown to order correctly experimental melting temperatures and can be used to predict melting temperatures with considerable accuracy. Our equivalence parameter was obtained by using the Peyrard–Bishop model to calculate maxima in the diagonal components of the partition function. As such a calculation is several orders of magnitude faster than the complete temperature calculation, we are now in a position to search for more optimized parameters.

The basic procedure for optimizing the model parameters is minimizing the square difference χ^2 between the predicted and measured melting temperatures, T_i' and T_i, for a given set of sequences (see also Supplementary Methods). We carried out the multidimensional minimization of χ^2 in two steps. First, we minimize χ^2 using the regression method with a logarithmic dependence for the salt concentration (see Supplementary Equations (S1)–(S3)). Once a satisfactory minimization of χ^2 has been achieved, we relax the restriction of the logarithmic salt-concentration dependence, Supplementary Equation (S3), and minimize χ^2 for each salt concentration independently. To estimate the numerical error of our minimization and to avoid local minima, we used a bootstrapping method where we change the melting temperatures by small random differences and re-run the minimization (see also Supplementary Methods).

We carried out the minimization for the melting temperatures reported by Owczarzy and colleagues[17]. We were able to reduce the average difference between experimental and predicted

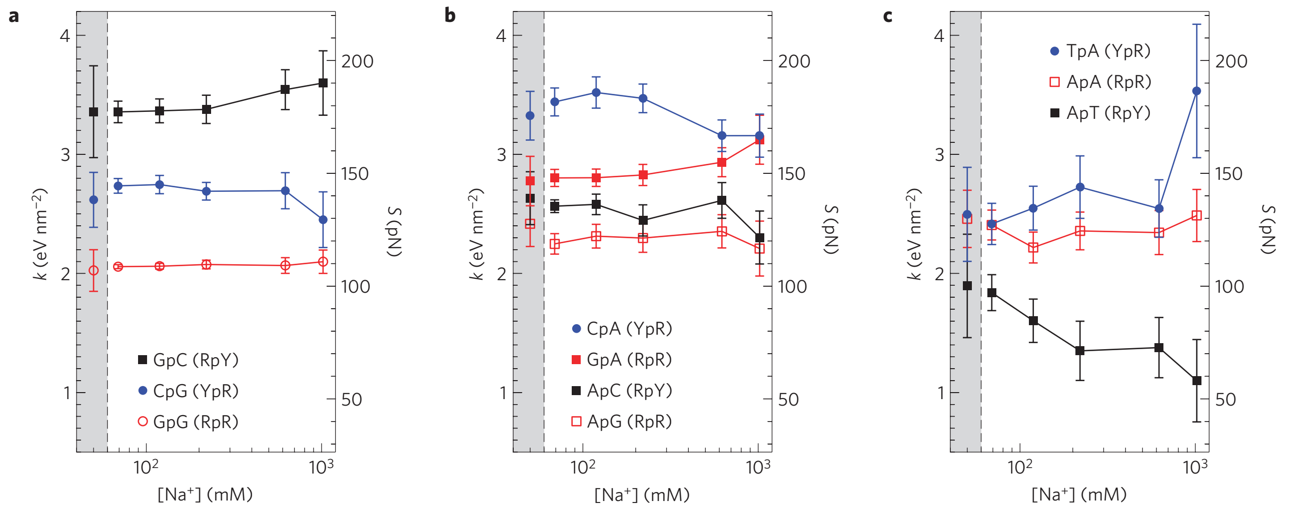

Figure 2 | Elastic constant, *k*, as a function of salt concentration. The grey rectangles show the parameters obtained for the first step of the minimization procedure where we impose a logarithmic salt-concentration dependence given by Supplementary Equation (S3). These parameters are used as starting points for the second step where we minimize for each salt concentration independently. **a**–**c**, The dimer steps are grouped in order of decreasing hydrogen bonds: totals of six (**a**), five (**b**) and four (**c**) hydrogen bonds. The error bars were calculated using the bootstrap method over 500 modified data sets (see Supplementary Methods).

temperatures from 1.39 to 0.83 °C, an important improvement if one considers the reported experimental uncertainly of 0.3 °C (ref. 17). The total processing time was 2,500 h or 105 days on 2 GHz processors. Similar minimizations were also carried out for different Peyrard–Bishop Hamiltonians with anharmonic stacking potentials[24] and with Morse-solvent potentials[25] (data not shown). However, we have found no substantial improvement in the melting temperature predictions when using either of these Hamiltonians.

Figure 1 shows the results for the base-pair parameters D and λ, which are related to the hydrogen bonds. Basically, there is no significant salt dependence for these parameters. The final values for the Morse parameters are very close to the initial parameters obtained in early studies by Campa and Giansanti[26]. The Morse potentials D obtained from the multidimensional minimization presented in Fig. 1 are consistent with those found in a previous work where we minimized χ^2 by simply varying uniformly the value of D (Fig. 5 of ref. 15). The larger error bars for the A · T λ parameter simply reflect the fact that the Morse potential becomes shallower for larger λ (see also Supplementary Information and Supplementary Fig. S1).

The parameters related to DNA flexibility are shown in Fig. 2, which shows the salt dependence of the elastic constants for all ten dimer steps, that is, all conformations of two consecutive base-pair steps. These steps can be characterized according to nucleotide groups of purines R (G and A) and pyrimidines Y (C and T). The elastic constants of purine–purine dimer steps (RpR), which are equivalent to pyrimidine–pyrimidine dimer steps (YpY), are essentially left unchanged with increasing salt concentration. Important changes however are observed for pyrimidine–purine (YpR) and purine–pyrimidine steps (RpY), especially for weakly bonded dimers. Furthermore, YpR and RpY show generally opposite trends with increasing salt concentration. ApT and TpA show the strongest variation of the elastic constant; at high salt concentration, TpA becomes very stiff, whereas ApT becomes extremely soft. These extreme opposite trends in elasticity with salt concentration are consistent with non-uniform charge distribution mediated by counter-ions[27–30]; such non-uniform charge distribution does not occur for the RpR dimer steps.

The elastic constants in Fig. 2 are also shown as stretching moduli S, that is, the force required to double the length of the polymer. The orders of magnitude of the stretching moduli found are in agreement with the value of 91 pN recently measured by Mathew-Fenn and colleagues[6]. In general, our results are also in line with several recent findings that DNA is much more flexible on short scales than on long scales[4,5,11]. Our results differ from established flexibility properties of protein-bound DNA where YpR steps generally have the highest flexibility and RpY steps the lowest[8]. In our case the flexibilities are much more dominated by hydrogen bonds, which is not surprising because they were obtained from temperature measurements. On the other hand, protein-bound DNA measurements use six degrees of freedom to describe the flexibility of DNA, which makes direct comparisons with our results quite difficult[8]. Figure 2 also reveals that the two main contributors to Na^+ salt-dependent flexibility are TpA and ApT dimers, which show a completely different behaviour from all other dimer steps. This result can be explained by preferential A-tract binding of Na^+ ions as well as by the fact that these bind very differently to TpA and ApT dimers[27–30]. Consequently, TpA and ApT dimers stand out as major contributors to changes in Na^+ salt-dependent melting temperatures.

One important and well-known flexibility property of genomic DNA is the presence of TATA regions or boxes, which suffer severe bending when linked to TATA-box binding protein[18,20]. TATA-box binding protein is responsible for transcription initiation and its specificity is influenced by the fact that TATA regions are located in very flexible genomic regions[18–21]. Marilley *et al.*[7], for instance used atomic force measurements to probe the flexibility of a DNA molecule and were able to show the distinctively higher flexibility near the TATA region. In Fig. 3 and Supplementary Fig. S2, we show some examples of flexibilities in promoter regions and TATA boxes. To have an estimate of the flexibility of the genomic DNA, we calculated the equivalent elastic constants considering windows of four base pairs, a sufficiently short scale where nonlinear effects are probably of minor importance. In this case, the elastic constants were calculated as classical coupled elastic springs from Supplementary Equation (S5). The TATA-box region in all of these cases is one of the most flexible in this genomic neighbourhood.

The method presented here provides a way to look at existing melting temperature data to obtain new information about the flexibility of DNA in solution. Several of the well-known properties of DNA such as the high flexibility of TATA regions were obtained. The good quantitative agreement of our

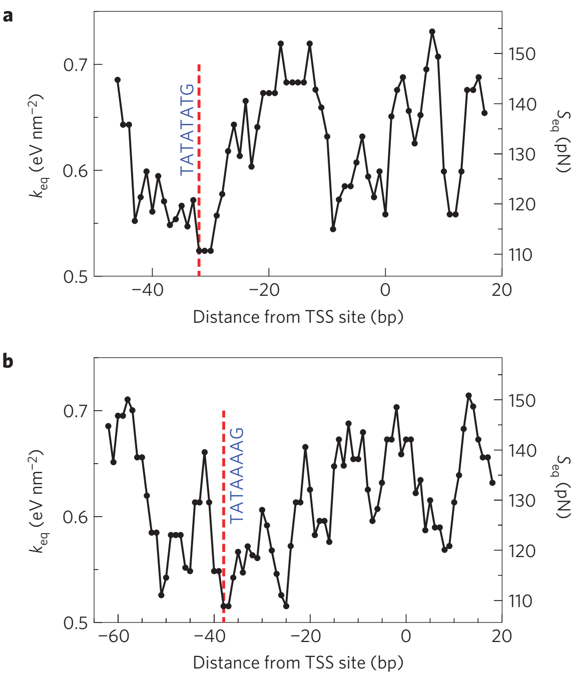

Figure 3 | Equivalent elastic constants, k_{eq}, for promoter regions. The salt concentration is 69 mM for all cases, and calculations were carried out with a window of $N = 4$ bp; see Supplementary Equation (S5). **a**,**b**, TATA-box regions (red dashed line) are shown for a p5 viral promoter[34] (**a**) and an AdMLP promoter[34] (**b**).

results with current knowledge on flexibility properties for very short sequences is an indication that the Peyrard–Bishop model can provide a good description of DNA flexibility. Therefore, this model now provides a basis for using nonlinear dynamic models to describe nanoscale flexibility in DNA. Furthermore, we now finally have a detailed description of the Na^+ dependence of DNA flexibility at the nanoscale. These details reveal that TpA and ApT dimers are mainly responsible for changes in DNA flexibility as well as melting temperatures with varying Na^+ concentration. In addition, these findings are consistent with preferential ion binding to TpA and ApT dimers observed from molecular dynamics[27,29], electrophoresis[28] and NMR-based structural modelling[30] approaches.

One limitation of the method is that it requires a relatively large number of sequences to obtain results with sufficient reliability. However, measuring the melting temperature of oligonucleotides is sufficiently straightforward, and specific measurements could be designed aiming for flexibility properties under varying solvent conditions. Although understanding the flexibility of DNA is very important, the applications of our results go beyond this: the Peyrard–Bishop model was proposed 20 years ago, yet our knowledge of the parameters required for its use has always been very limited. Therefore, the flexibility results presented here provide a much needed microscopic validation for this class of models. Our results will enable us to apply this important approach to a much larger range of phenomena related to DNA and perhaps to other oligonucleotides and may even help to improve the model itself. The multidimensional minimization method described here can be used to validate modifications to the model such as the addition of a helicoidal Hamiltonian[31].

Traditionally the flexibility of DNA on a large, that is, mesoscopic scale has been described by means of a persistence length obtained from the WLC model[9,10] or some of its variations[32]. However, establishing a connection between microscopic and mesoscopic flexibilities is not straightforward. The one-dimensional Peyrard–Bishop model[13] that we use in this work lacks the angle correlation between adjacent base pairs necessary for an explicit calculation of persistence lengths. This limitation could be resolved for instance with the addition of a helicoidal degree of freedom[31] to the Hamiltonian in equation (1). Alternatively, it should be possible to use the elastic constants of Fig. 2 in combination with a breathing WLC model that takes into account the dynamic bubble formation in the DNA chain[33]. However, in our view the ability to describe microscopic flexibility supersedes the use of such long-range averages. By using data from non-perturbing solution experiments to determine the flexibility of specific sequence elements, our approach makes it possible to analyse flexibility base by base, enabling us to probe changes and variation at the level at which selective protein–DNA and DNA–DNA interactions occur—that is, at the level at which the biology happens.

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Author contributions

G.W. conceived the method and carried out the calculation. C.N. and J.W.E. contributed with the biochemical interpretation of the data and gave conceptual advice. All authors wrote the paper.

Additional information

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